Oxidation of Metals, Vol. 53, Nos. 1/2, 2000

The Influence of the Alloy Microstructure on the Oxidation Behavior of Ti–46Al–1Cr–0.2Si Alloy

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Received October 13, 1998; revised February 23, 1999

The influence of microstructure of the two-phase alloy Ti–46Al–1Cr–0.2Si on the oxidation behavior in air between 600 and 900°C was studied. The oxidation rate, type of scale, and scale spallation resistance were strongly affected by the type of microstructure, i.e., lamellar in as-cast material and duplex after extrusion at 1300°C. The oxidation rate was affected by the size and distribution of the α_2-Ti_3Al phase, being faster for the extruded material with coarse α_2-Ti_3Al. The type of oxide scale determines the spalling resistance. Cast material developed a uniform scale that spalled off after short exposure times at 800 and 900°C when a critical thickness was reached. The extruded material presented a heterogeneous scale with predominant thick regions formed on γ-TiAl–α_2-Ti_3Al grains and thin scale regions formed on γ-TiAl grains. This type of scale could permit an easier relaxation in the matrix of stresses generated by both thermal-expansion mismatch between scale and alloy and oxide growth, resulting in a higher spallation resistance.

KEY WORDS: γ-TiAl-based intermetallic alloy; oxidation; microstructure.

INTRODUCTION

The large research effort realized by the aircraft industry in the development of structural materials for high-temperature application has led to the consideration of γ-TiAl-base alloys because of their high specific strength and

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high Young's modulus. However, low ductility at room temperature (2–3%) and poor oxidation resistance at temperatures higher than 700°C have limited their use. Modification of binary alloys by ternary and multielement additions leads to improvement of both properties. Elements such as V, Mn, and Cr generally increase the ductility,[1] while the oxidation behavior can be improved by alloying additions such as Mo, Nb, Hf, Zr, Cl, or W.[2–18] However, each of these alloying elements benefits only a specific property and is generally detrimental for the other. It is difficult to achieve an equilibrium between good mechanical properties and oxidation resistance in γ-TiAl base alloys only by alloying elements.

Most of the work on the oxidation behavior of γ-TiAl base alloys has been directed toward the determination of the effect of alloying elements[5–18] and/or the oxidation mechanism.[8,19–29] Less attention, however, has been devoted to the influence of the alloy microstructure on the oxidation behavior.[30–37] This aspect can be very important in the most promising γ-TiAl base alloys since the small volume fraction of α_2-Ti_3Al allows manufacturing of several microstructures, i.e., duplex or fully lamellar. A duplex microstructure consists of both γ-grains and grains with a lamellar structure. The lamellar grains contain alternating γ and α_2 plates. The grains are typically 10–25 μm in size and the lamellar plates are 0.1–1 μm thick. Finally, alloys treated in the single-phase α field exhibit the fully lamellar structure with grains typically greater than 500 μm in size. The different distribution and volume fraction of the second phase in each microstructure can affect both mechanical properties and oxidation resistance.

Studies about the effect of type microstructure on the oxidation behavior report contradictory results. Gil *et al.*[30,31] reported for a two-phase γ/α_2-TiAl alloy oxidized at 900°C in air that the oxidation rate and type of the scale developed depend upon the type of microstructure and distribution of both phases. Cast material, with a fine lamellar microstructure, formed a thin protective scale instead of the thicker rutile-base scale established on heat-treated material, whose microstructure consisted of large grains of γ-TiAl and coarse α-Ti_3Al. A similar result was reported by Brady *et al.*[33,34] for cast and heat-treated two-phase (γ-TiAl and Laves phase) Ti–45Al–15Cr alloy. The coarser microstructure on the heat-treated alloy resulted in a depth attack in γ-TiAl regions. However, Shemet *et al.*,[25] for TiAl oxidized in argon–oxygen at 900°C, and Haanappel *et al.*,[36,37] for Ti–48Al–2Cr oxidized in air at 700 and 800°C, reported that oxidation rate was unaffected by the alloy microstructure.

In the present work, the oxidation behavior of both as-cast and extruded Ti–46Al–1Cr–0.2Si alloys has been studied to elucidate the influence of the type of microstructure. The composition of the alloy was selected

following a search for a material with both high mechanical properties and sufficiently good oxidation resistance.

EXPERIMENTAL PROCEDURES

Chromium-alloyed two-phase γ/α_2-TiAl alloy was produced in 50-kg ingots by arc skull melting. The chemical composition of cast material is Ti 52.8–Al 46–Cr 1–Si 0.19 (at.%). The microstructure of as-cast material was refined by hot extrusion of billets 75 mm in diameter and 120 mm in length in the $\alpha + \gamma$ field at 1300°C. The reduction in area of the extruded rods was about 7:1.

Oxidation tests were carried out on rectangular coupons of $10 \times 10 \times 1\ \text{mm}^3$ cut from the billets by electrospark erosion. All major surfaces were abraded on successively finer silicon carbide papers, then mechanically polished with 1-μm diamond paste, and finally ultrasonically cleaned with acetone. To determine the oxidation kinetics, a discontinuous thermogravimetry technique was used in the temperature range of 600 to 900°C. To verify the actual test temperature, a thermocouple was installed close to the sample. After the exposure, samples were left to cool out of the furnace and weighed in a balance with a resolution of $\pm 10\ \mu$g.

Characterization of oxidation products was performed on samples isothermally oxidized for short- and intermediate-exposure times, as well as on samples used for the oxidation kinetics determination. To prevent scale loss during metallographic preparation, sample surfaces were successively coated with a thin gold layer (by sputtering) and a thicker layer of copper (electrolytically deposited). Cross sections were prepared by conventional metallographical techniques, without alumina-grinding media. Surfaces and cross sections of the oxidized specimens were studied by scanning electron microscopy (SEM). Phase identification of the oxide scale was performed by X-ray diffraction (XRD), energy-dispersive X-ray microanalysis (EDX), and wavelength-dispersive X-ray microanalysis (WDX).

RESULTS

Microstructure

The microstructure of both as-cast and as-extruded materials is presented in Fig. 1. The material in the as-cast condition shows coarse grains ranging from 100 to 200 μm with an alternating arrangement of plaques of γ-TiAl and α-Ti_3Al (Fig. 1a). The volume fraction of α-Ti_3Al phase was between 15 and 20%. This lamellar structure was partially transformed by extrusion at 1300°C into a duplex microstructure consisting of γ

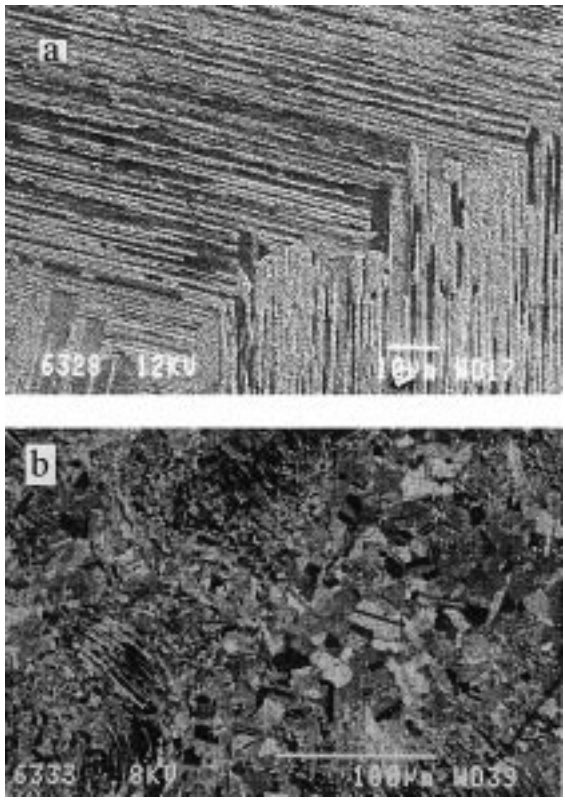


Fig. 1. Surface-scale morphology after 160 hr oxidation at 600°C showing microstructure of (a) as-cast Ti–46Al–1Cr–0.2Si alloy and (b) extruded Ti–46Al–1Cr–0.2Si alloy (SEM micrographs).

single-phase grains ranging from 5 to 20 μm and lamellar γ/α_2 two-phase grains, as shown in Fig. 1b. The thickness of the α_2 lamellar plaques was larger than for the as-cast material. Observations in the extrusion direction show that both γ single-phase and lamellar grains were grouped in alternating arranged bands ranging from 60 to 150 μm in width. EDX measurements indicated a slight Cr enrichment of γ/α_2 duplex regions compared to γ single-phase regions

Oxidation Kinetics

Figure 2 shows the mass-gain–time curves for both extruded and as-cast material. At 600°C, the mass gains were negligible and the curves for

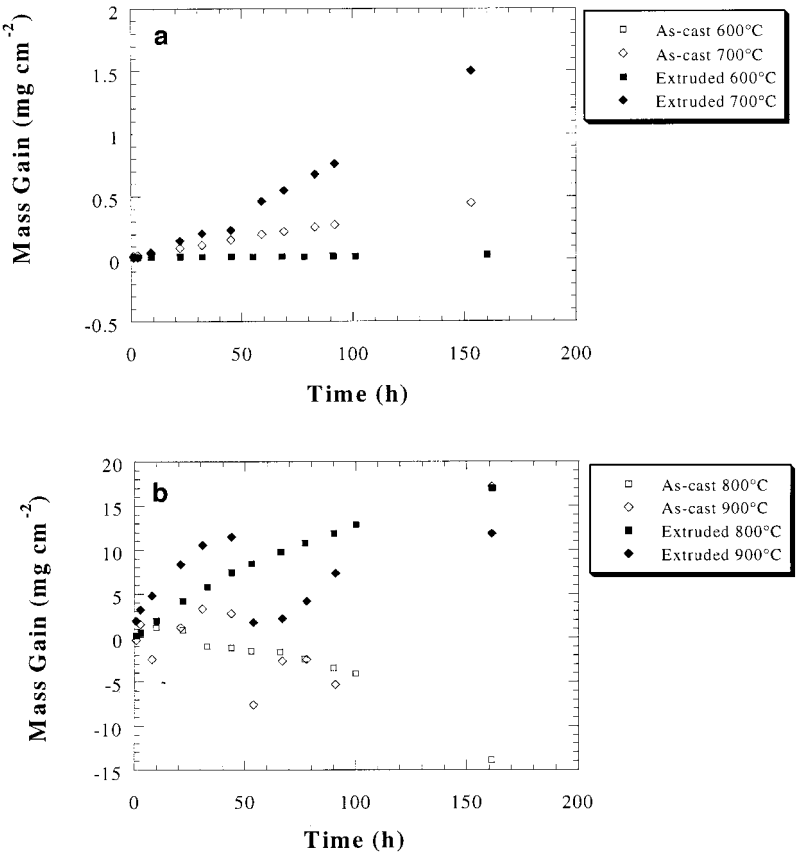


Fig. 2. Oxidation kinetics of both as-cast and extruded Ti–46Al–1Cr–0.2Si alloy oxidized in air. (a) 600 and 700°C; (b) 800 and 900°C.

both materials can be superimposed. At 700°C, the oxidation rate for both alloys was practically the same during the early stages of oxidation. However, the extruded material experienced a more significant mass gain with progressing exposure time with respect to the as-cast material. At temperatures above 700°C, the oxidation rate for both materials was significantly accelerated, with higher mass gains always for the extruded material. At 800°C, the as-cast material exhibited a mass loss after short exposure times, whereas the extruded material had a continuous mass gain. At 900°C, spallation of the scale produced a mass loss after 5 hr for as-cast and after 45 hr for the extruded material. For longer exposure times, mass-gain and mass-loss periods can be observed.

Surface Morphology

The thin scale formed after testing at 600°C revealed the microstructure of both alloys (see Fig. 1), in which the dark and bright zones are formed over the γ-TiAl and α_2-Ti_3Al phases respectively. EDX analysis of both zones revealed that dark zones were constituted by an Al-rich oxide (likely alumina), while bright zones consisted of an Al-rich oxide covered by a very porous and discontinuous Ti-rich oxide (probably rutile). At grain boundaries of the extruded material, small Si-rich precipitates (presumably silica) were also found.

At 700°C, a much thicker scale consisting of a mixture of Al and Ti oxides formed at the outer surface. This scale presented a slightly convoluted aspect that just allowed characterization of the substrate microstructure. For the longest exposure, well-developed rutile crystals were observed only in the extruded material, as shown in Fig. 3. This suggests that outward diffusion of Ti through the scale was slower in the case of as-cast material. The presence of rutile crystals seems to account for the higher mass gain measured for the extruded material.

At 800 and 900°C, the morphology of the scale of both materials was completely different from that described before. In both cases, the outer layer, even from the earlier stages of oxidation, consisted of the rutile phase. The size of prismatic rutile crystals increased with increasing temperature and exposure time, up to 100 μm after 160 hr at 900°C, as presented in Fig. 4. Although both materials developed an outer rutile scale, some differences concerning the morphology of the scales were found. The as-cast material formed a homogeneous smooth scale for all exposure times (Fig. 5a). At places where spallation occurred, the scale was composed of a mixture of Al and Ti oxides. The scale on the extruded material, however, showed for short times a convoluted aspect with regions of different thickness (Fig. 5b). The rutile crystal size was smaller in the thin regions. For longer times, the scale exhibited nodules (see Fig. 6) with a similar composition and morphology to the rest of the scale.

Cross Sections

Cross-sectional views of the oxidized samples also showed differences in the oxide scale, related mainly to the uniformity in thickness between the as-cast and extruded material. As shown in Fig. 7, the as-cast alloy developed a thin and homogeneous oxide scale, whereas the extruded alloy presented a thick and heterogeneous scale.

At 700°C, the scale formed on the as-cast material consisted of three layers containing a mixture of rutile and alumina (Fig. 8a). Both the outer and inner layers were alumina-rich, whereas the intermediate layer was

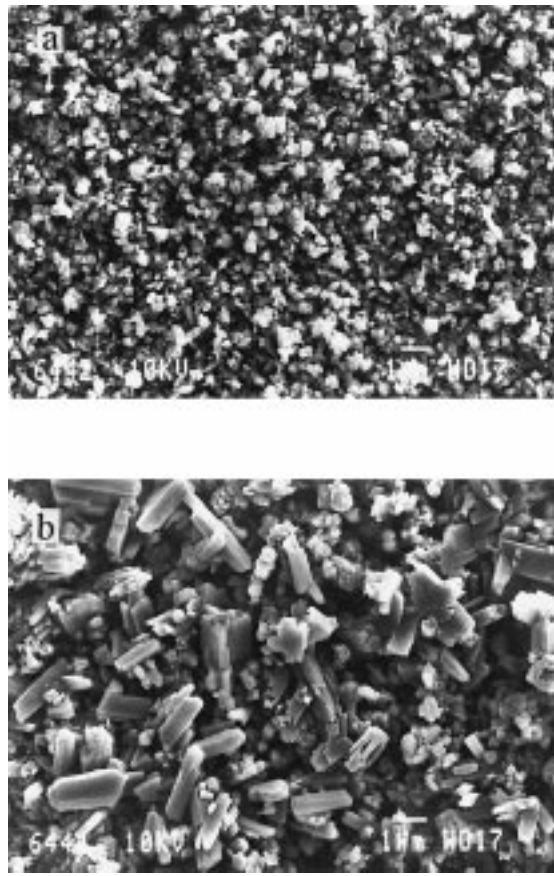


Fig. 3. SEM micrograph showing surface scale morphology after 150 hr at 700°C. (a) As-cast Ti–46Al–1Cr–0.2Si alloy; (b) Extruded Ti–46Al–1Cr–0.2Si alloy.

rutile-rich. At the oxide–alloy interface, a discontinuous band of a Ti-rich phase was present. The scale formed in the extruded material was more complex. It was possible to distinguish predominantly thick-scale regions of about 10 μm connected by regions in which the scale was thinner (around 2 μm). In both regions, the scale pattern found was similar, although with varying thickness of the different layers. Four layers forming the oxide scale were recognized, as presented in Fig. 8b. The outermost layer consisted mainly of an Al-rich oxide for the initial stages up to 50 hr and became a rutile-rich layer with longer exposure times. This layer was followed by a thin alumina-rich barrier and a thicker, inner-intermediate layer containing

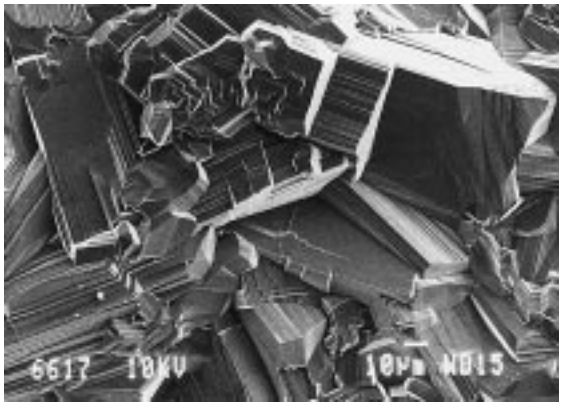


Fig. 4. Rutile crystals formed on the surface of the scale after 160 hr at 900°C (SEM micrograph).

alumina (dark phase) and rutile (bright phase). The thin innermost layer, close to substrate, was composed mainly of alumina. Chromium was detected in alumina-containing zones, the concentration increasing toward the scale–substrate interface. An Al-depleted zone was detected in the alloy beneath the scale–alloy interface.

At 800°C, a thicker scale with an outer layer constituted exclusively of rutile was observed on both alloys. For the as-cast material, the thickness of the oxide scale was 5 and 22 μm after 10 and 160 hr, respectively. As shown in Fig. 9a, this scale consisted of a thin outer-rutile scale, an intermediate alumina-rich layer, and an inner thicker alumina–rutile mixed layer, whose external part corresponded to nearly pure rutile. For longer times, a new inner alumina-rich layer appeared at the scale–metal interface (Fig. 9b). At this scale–metal interface, a thin discontinuous band of a Ti-rich phase was found after 10 hr. As can be deduced from the XRD pattern of Fig. 10, this phase corresponded to titanium nitrides, Ti_2AlN and TiN. For the longest exposure time of 160 hr, this phase was not observed, indicating that it could originate during the initial stages of oxidation. Although every layer in the scale thickened as oxidation proceeded, the fastest growth was experienced by the mixed rutile-alumina layer. In this mixed layer, the lamellar microstructure of the alloy was recognized by etching, as shown in Fig. 11. Instead of the Ti-nitrides, a thicker two-phase Al-depleted zone of about 10 μm was found at the scale–metal interface in which both phases were arranged with the orientation of the original γ and α_2 lamellae. EDX analysis of the Al-depleted zone in the as-cast alloy revealed that this layer contained a large amount of oxygen (Table I).

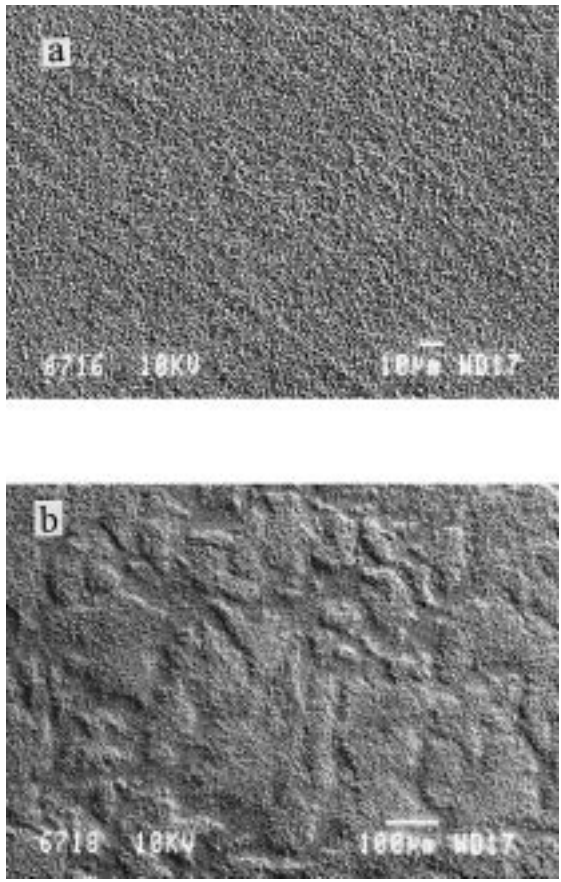


Fig. 5. SEM micrographs of the surface scale formed after 10 hr at 800°C. (a) As-cast alloy; (b) extruded alloy.

After 10 hr of exposure at 800°C the extruded material showed two scale regions, a thicker (around 15 μm) and a thin (3–4 μm) scale (see Fig. 7d). In both regions, the scale was apparently formed by four layers disposed in the same sequence as that observed in the as-cast material after longer times (Fig. 12). Again, at the scale–metal interface an Al-depleted zone was observed. Etched samples revealed that thicker and thinner scale were formed over the lamellar γ/α_2 two-phase and γ single-phase grains, respectively (see Fig. 13). Unlike the two-phase Al-depleted zone developed beneath the thick scale, a single-phase zone was observed beneath the thinner scale. After 160 hr, the air–scale interface also showed a very irregular

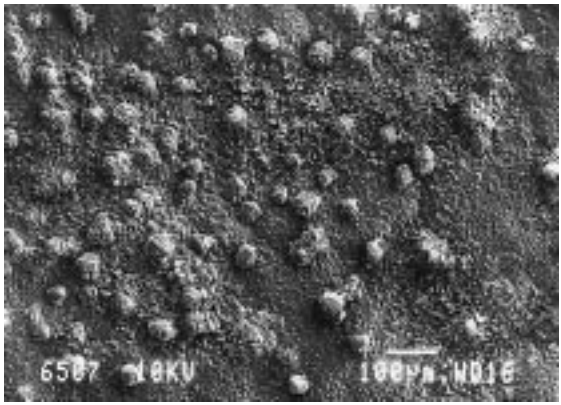


Fig. 6. Outer scale formed on extruded alloy after 160 hr at 800°C (SEM micrograph).

profile. The scale was composed of a thick outer rutile layer (up to 50 μm) with a large density of nodules, consisting of alumina at the core and rutile at the external part (Fig. 14). Beneath the outer rutile scale appeared a nearly continuous thin alumina-rich layer, followed by a thin rutile layer. The thickest layer of the scale, about 60 μm, was presented by the inner mixed rutile–alumina layer, in which the part close to the scale–alloy interface displayed alternating rutile-rich and alumina-rich bands. Chromium was also found in this mixed layer with a concentration similar to that of the alloy. Beneath the scale, a two-layered Al-depleted subsurface zone containing a large amount of oxygen was present (see Table I).

At 900°C, the scale formed for short times corresponded well to the one observed for intermediate times at 800°C, as presented in Fig. 15. After 1 hr, a scale of 8 μm and between 6 and 13 μm was found for the as-cast and extruded materials, respectively. These scales evolved for longer times to a uniform scale up to 350 μm and to a nonuniform scale with a maximum thickness of 300 μm for the as-cast and extruded materials, respectively, as observed in Fig. 16. The scale formed in as-cast material consisted of an outer thick rutile scale with some dispersed alumina, a discontinuous alumina barrier, a rutile-rich layer, an inner mixed layer with an alternating arrangement of alumina and rutile sublayers, and an Al-depleted subsurface. EDX microanalysis of etched samples indicated that the subsurface layer was composed of two oxygen containing phases with the orientation of α_2 plaques (see Fig. 17 and Table II). Prominent grains corresponded to oxygen saturated α_2-Ti_3Al, whereas depressed grains corresponded to $Ti_5Al_3O_2$. As it is observed in Fig. 17, the oxidation progressed preferentially in the Al-depleted zone through the prominent grains, i.e., α_2-Ti_3Al phase,

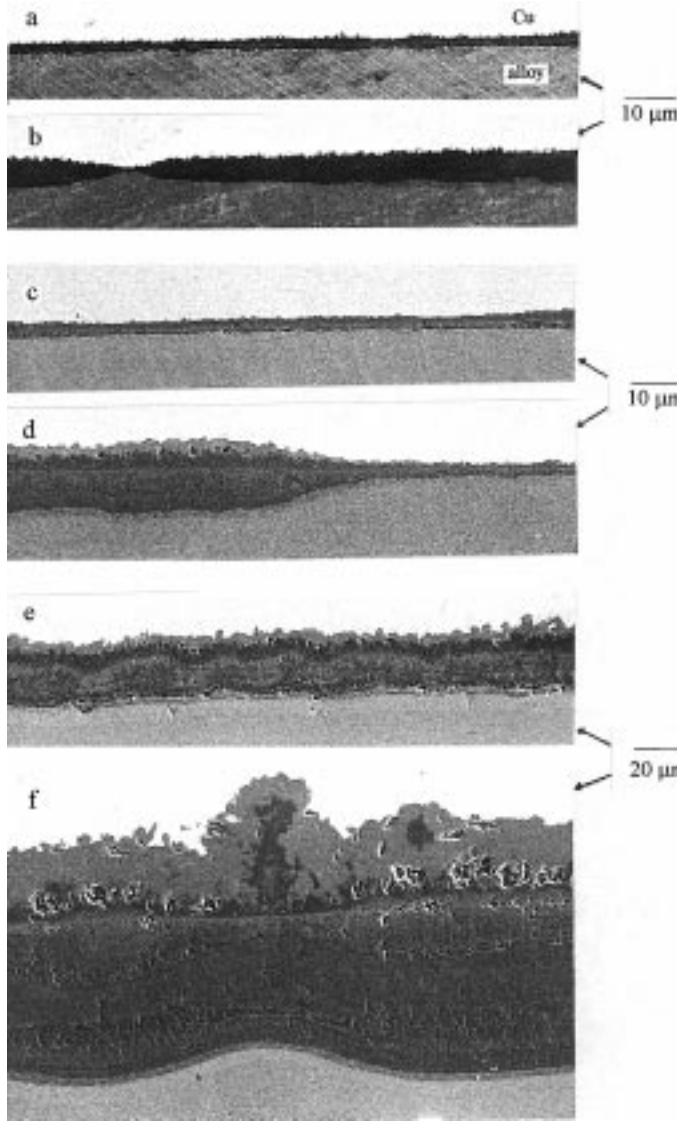


Fig. 7. Cross sections of scales formed in as-cast (a, c, e) and extruded alloy (b, d ,f). (a, b) After 150 hr at 700°C, (c, d) 10 hr at 800°C, and (e, f) 160 hr at 800°C (e, f) (SEM micrographs).

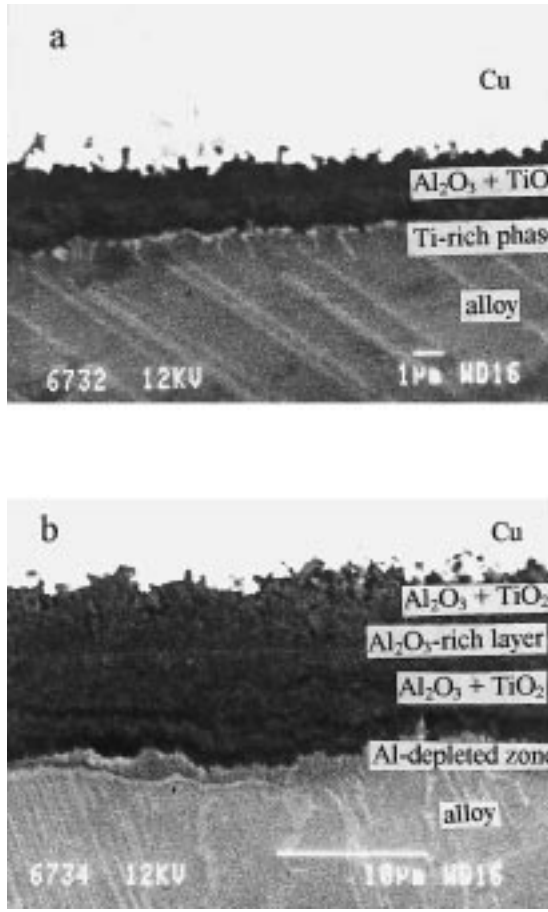


Fig. 8. Cross sections of scales formed after 150 hr at 700°C. (a) As-cast alloy; (b) extruded alloy (SEM micrographs).

indicating that oxidation resistance of this phase is less than that of $Ti_5Al_3O_2$.

For the extruded material, the structure of the scale is similar to that found in as-cast material, although the scale–alloy and, especially, the scale–gas interfaces were very rough. The outer rutile scale was characterized by formation of large protuberances, up to 200 μm in thickness (see Fig. 16b). The alumina barrier was also more compact and the inner mixed alumina–rutile layer was thinner than in the as-cast material.

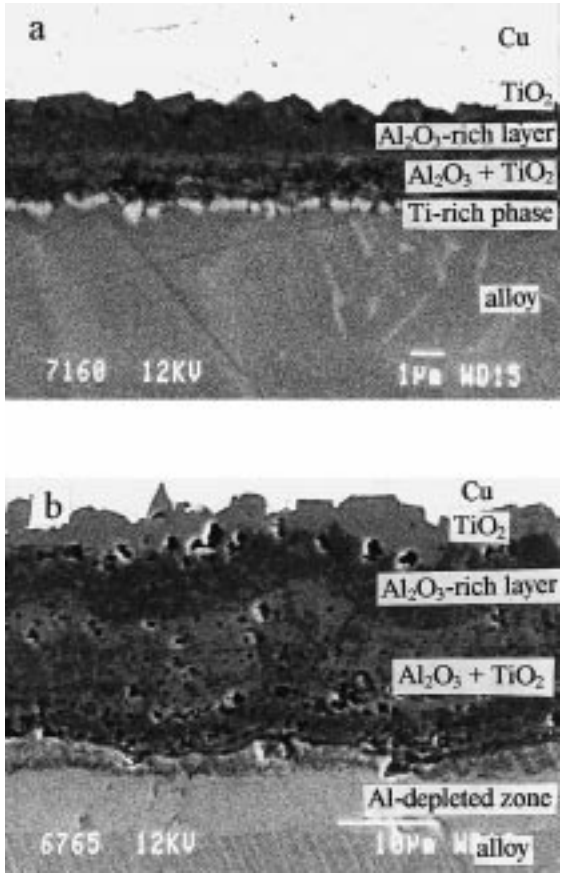


Fig. 9. Cross sections of the as-cast alloy oxidized at 800°C. (a) 10 hr; (b) 160 hr (SEM micrographs).

DISCUSSION

It is already well known that oxidation of TiAl can be divided in several stages.[2,3,22,24,26,27] An increase of temperature compresses these stages to shorter times, because each stage is affected by different diffusion rates in the scale and substrate.[26] As this behavior has been found in the present work, the results will be discussed separately for each temperature.

The results obtained in this work show that the oxidation rate, morphology, and thickness of scale and scale-spallation resistance depend on the type of microstructure of the alloys as well as on the oxidation

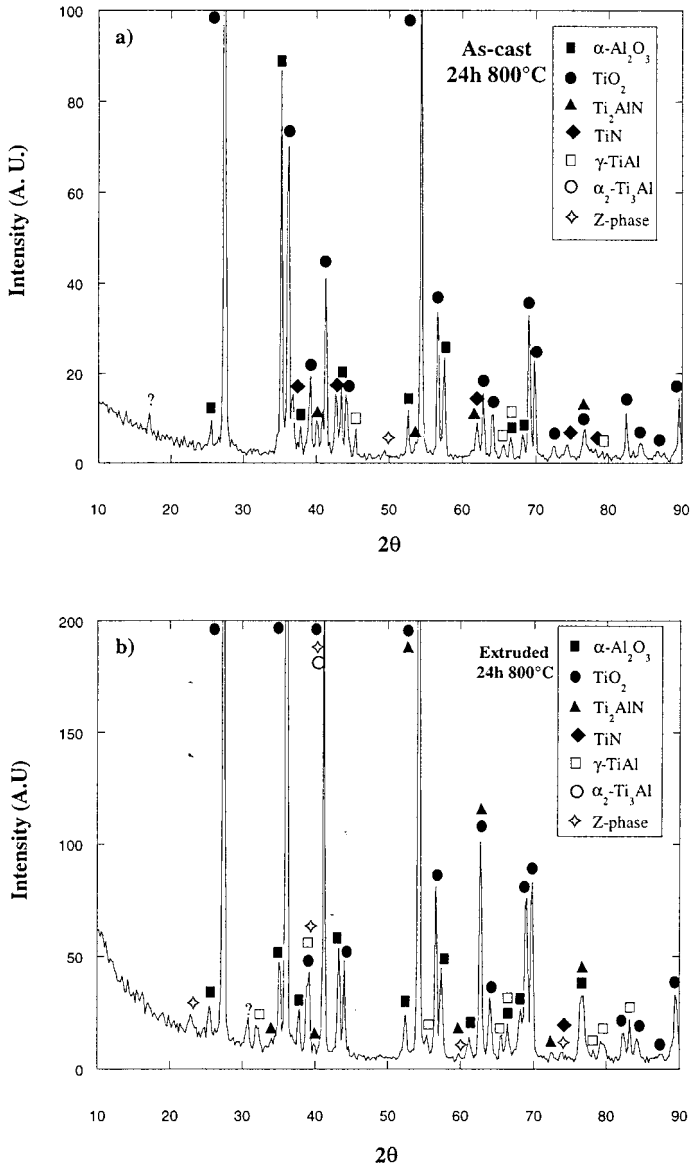


Fig. 10. XRD patterns after 24-hr oxidation at 800°C. (a) As-cast alloy; (b) extruded alloy.

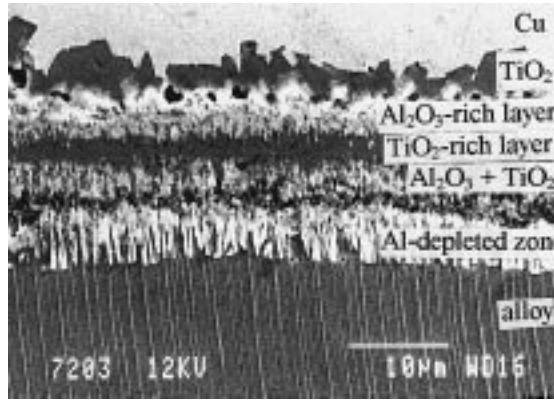


Fig. 11. Etched cross section of the as-cast alloy oxidized for 160 hr at 800°C (SEM micrograph).

Table I. Composition of the Al-Depleted Zone, Analyzed by EDX, in As-Cast and Extruded Alloys After 160 hr Oxidation at 800°C

Alloy	Ti (%)	Al (%)	Cr (%)	O (%)
As-cast	53.4	27.8	—	18.8
Extruded	55	30.5	0.5	14

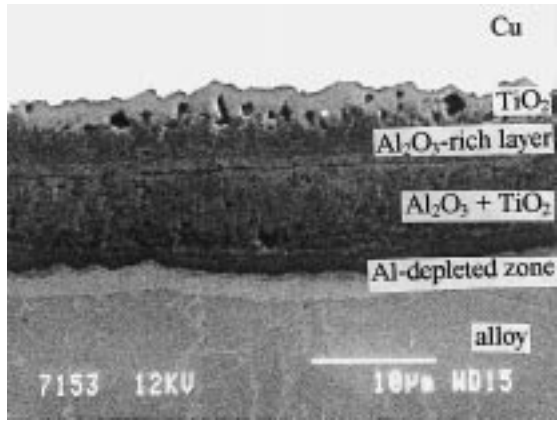


Fig. 12. Cross section of the extruded alloy oxidized 10 hr at 800°C (SEM micrograph).

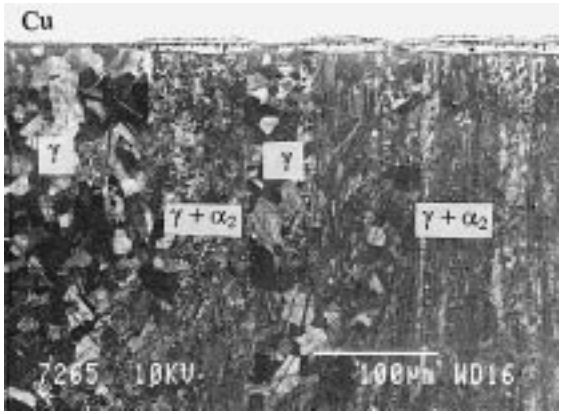


Fig. 13. Etched cross-section of the extruded alloy oxidized 10 hr at 800°C (SEM micrograph).

temperature. At 600°C, similar low-mass gains were experienced for both the cast and extruded alloy, indicating that the type of microstructure does not affect oxidation behavior. The presence of rutile, probably formed during the initial stages of oxidation, over the alumina-rich film only on α_2-Ti_3Al lamellae, is an indication that at this temperature this phase forms transient Ti-oxides prior to establishing an alumina-rich layer while γ-TiAl does not. The slow oxygen diffusion in γ-TiAl and α_2-Ti_3Al would assist in establishing a continuous, protective alumina-rich film on both phases. As is well known, the high oxygen solubility of α_2-Ti_3Al[38] or Ti[39,40] alloys and the oxygen dissolution into the metal contribute significantly to the total mass gain in both materials.[40–43] Thus, two different processes give the total oxidation rate: oxide growth and oxygen dissolution into the metal. Over a particular temperature range, one process leads to significantly higher mass gain than does the other. The former process controls the oxidation in pure oxygen at temperatures below 650° and 700°C in pure Ti[39] and Ti–14Al–21Nb (at.%) alloy,[41] respectively, while inward oxygen transport controls oxidation for higher temperatures. In the case of the Ti_3Al-base alloy, the control of the oxide growth increases up to 800°C when oxidized in air because of the reduction of oxygen entrance by formation of a TiN layer.[41] Although the extruded material presented coarser α_2-Ti_3Al lamellae, the slower oxygen diffusion would permit the alumina-rich layer formed in γ-TiAl to undergrow the initial rutile layer developed on α_2 regions. This mechanism could be assisted by a local Al enrichment produced by two mechanisms: initial rutile formation and Al diffusion from the adjacent γ-TiAl, in a similar way to that suggested for alumina formation in an aligned lamellar eutectic Nb_3Al–Ni_3Al alloy.[44,45]

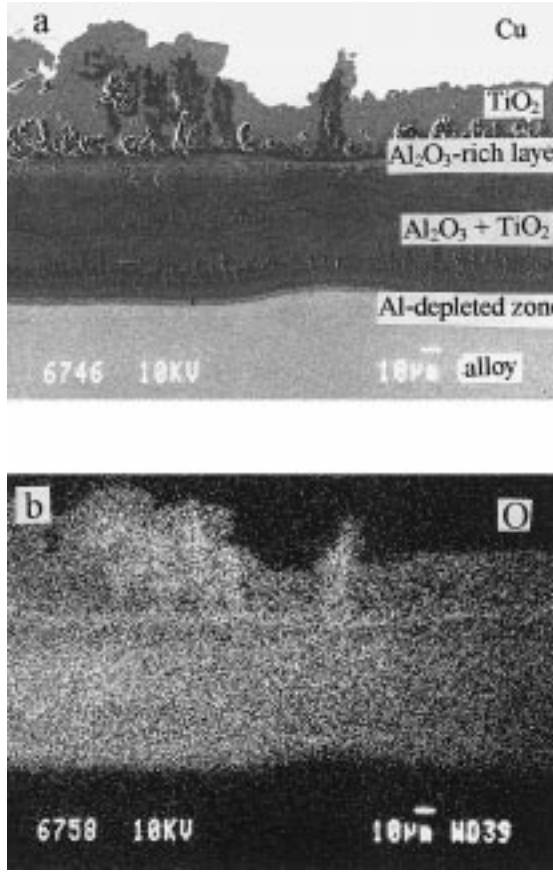


Fig. 14. Element-distribution maps for the extruded alloy oxidized 160 hr at 800°C: (a) SEM image; (b) oxygen; (c) titanium; (d) aluminum; and (e) chromium.

At 700°C, an alumina-rich film is formed from the early stages of oxidation. This layer remained protective during this period since a comparable lower oxidation rate was found for both types of microstructures. However, after 10 hr, the presence of rutile on the outer layer coincided with an increase in the oxidation rate, especially in the extruded alloy. The formation of this layer could imply outward Ti diffusion through α-alumina grain boundaries, as described in ODS Fe–Cr–Al alloys.[46,47] It is more probable that the large volume change produced by formation of rutile from the oxidation of inner α_2-Ti_3Al lamellae breaks up the outer alumina-rich film. This film impairment would be easier in extruded material due to the larger

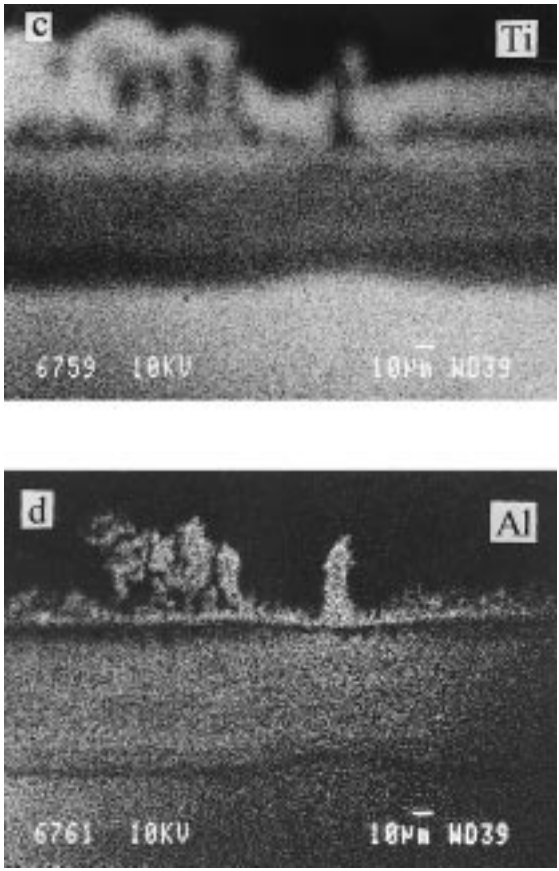


Fig. 14. Continued.

distortion introduced by oxidation of coarser α_2-Ti_3Al lamellae. The inward diffusion of oxygen through the oxide scale can also form an alumina-rich oxide when it reacts with the γ-TiAl phase, resulting in the development of a mixed rutile–alumina layer. Preferential growth of this mixed inner-oxide layer makes this layer the thickest part of the scale, as observed in the cross section.

TiN and Ti_2AlN were present at the scale–alloy interface of the as-cast alloy. However, this phase does not develop as a continuous layer that can decrease the oxidation rate. Moreover, Ti nitrides can lead to the inverse effect by preventing the establishment of a protective alumina layer.[2,28] For the extruded alloy, that formed a thicker scale, Ti nitrides were not detected.

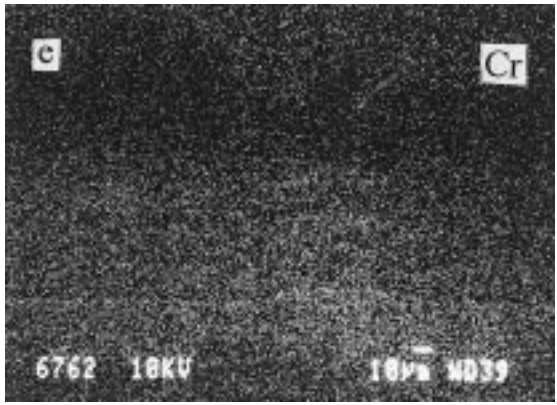


Fig. 14. Continued.

This suggests that nitride formation only took place during the initial stages and later disappeared when the scale reached a critical thickness. This agrees with the model proposed by Lang and Schütze[22,26] for γ-TiAl oxidation that considers only nitride formation during the initial stages of oxidation. These authors suggested that with increasing oxide thickness, the nitrogen partial pressure may drop below the equilibrium partial pressure of nitride formation and, consequently, it would be necessary to have a higher Ti activity to reform Ti nitrides. The present results are according to observations of Quadakkers *et al.*[8,21,23,27] and Meier *et al.*[48] who observed the disappearance of initially formed nitrides after some time.

The development of a uniform scale during oxidation was found only in the material with a homogeneous lamellar microstructure (cast alloy). For the extruded material, the thickness of the scale was very thin on γ-TiAl single-phase regions, evidencing the higher oxidation resistance of this phase. The duplex structure present in the extruded material leads to a heterogeneous scale with predominant thick regions, formed on γ/α_2 grains and thin scale regions, formed on γ-TiAl grains.

At 800°C, oxidation was already in an advanced stage after 10 hr, as denoted by the presence of a thin outer rutile layer. The scale formed practically coincided with that for longer times at 700°C, with a thick intermediate alumina barrier reducing the effective surface for oxygen diffusion. This alumina barrier does not prevent inward oxygen transport, as shown by the development of a thick inner rutile-alumina mixed layer. Outward Al diffusion in the scale to build up this alumina barrier has been explained by a decrease of alumina solubility in rutile when the oxygen partial pressure is increased.[20] As the oxide scale grows inward, the oxygen partial pressure

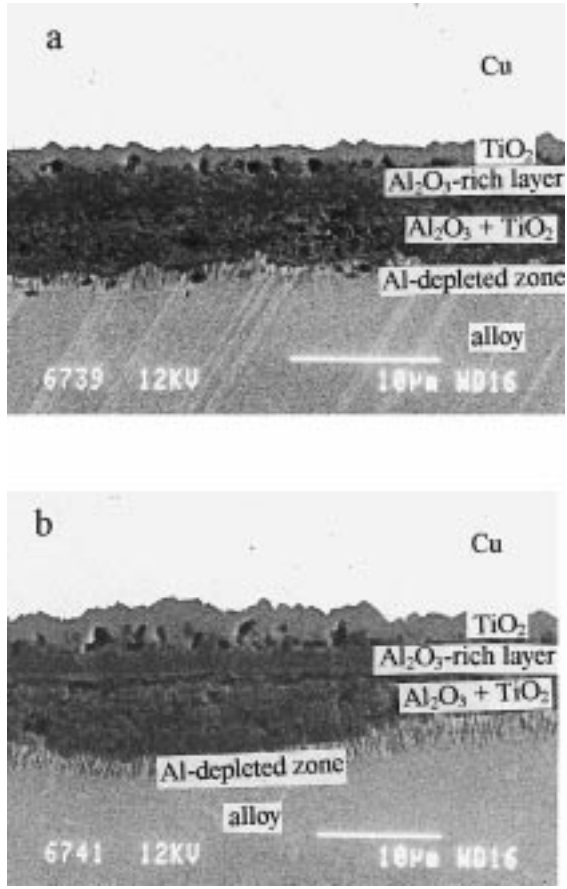


Fig. 15. Cross-sections of scales formed after 1 hr at 900°C. (a) As-cast alloy; (b) extruded alloy (SEM micrographs).

increases in the opposite direction, i.e., from the scale–alloy interface to the oxide surface. This gradient over the scale makes possible alumina dissolution in the inner scale at places where the oxygen partial pressure reaches a critical value and further re-precipitation at the intermediate alumina barrier takes place.[20] This mechanism would also explain the presence and thickening, with time, of a nearly pure rutile layer underlying the alumina barrier. For longer times, the more important changes occurred at the Al-depleted subsurface zone. Etched samples of fully lamellar material revealed that this zone consisted of two phases (see Fig. 11). On the other hand, etched extruded material revealed an Al-depleted zone with a single-phase and two-phase region underlying thin and thick scales, respectively.

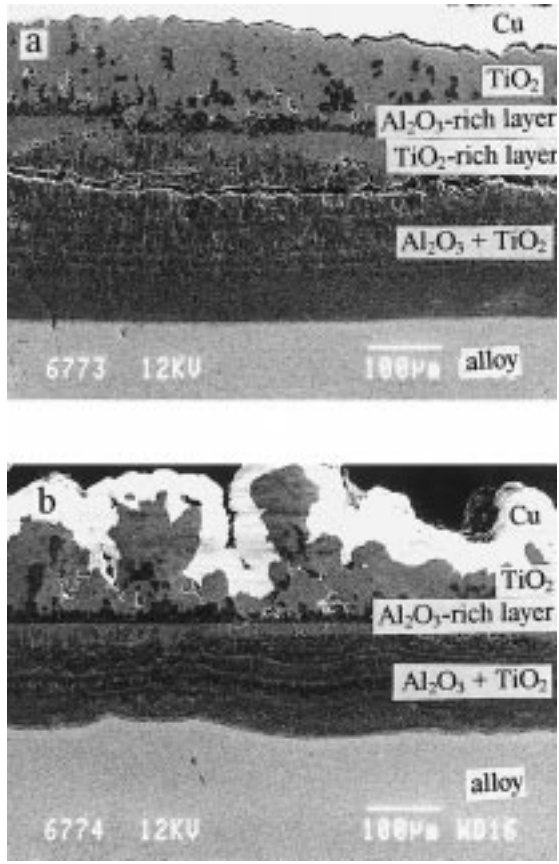


Fig. 16. Cross sections of scales formed after 160 hr at 900°C. (a) As-cast alloy; (b) extruded alloy (SEM micrographs).

Although the alumina barrier reduces the effective surface for oxygen diffusion, it cannot prevent oxygen entrance. The events occurring at the scale–alloy interface at this point might be important in further oxidation. As has been previously reported,[25–29] the processes that occur in this subsurface determine the oxidation behavior of γ-TiAl-base alloys in pure oxygen. In an early stage, a Ti–Al–O single-phase is developed at the Al-depleted zone, which has been designated in the literature as Z-phase,[23,25,49] X-phase,[24,28,49,50] NCP phase,[22,26] Ti_2Al[51] or $Ti_{10}Al_6O_2$ and $Ti_{10}Al_6O$.[52] The presence of a continuous Ti–Al–O compound in contact with the oxide scale guarantees a slow oxidation rate, because alumina is preferentially formed.

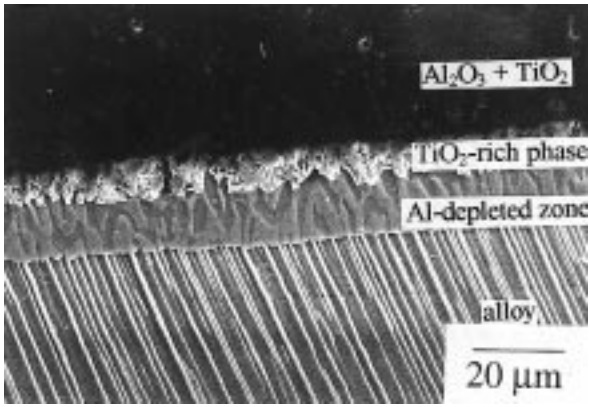


Fig. 17. Etched cross-section of the as-cast alloy oxidized for 160 hr at 900°C (SEM micrograph).

Table II. Composition of the Al-Depleted Zone, Analyzed by EDX, in As-Cast and Extruded Alloys After 160 hr Oxidation at 900°C

Alloy		Ti (%)	Al (%)	Cr (%)	O (%)
As-cast	Depressed grains	47.8	30.6	—	21.6
	Prominent grains	60.5	30.3	—	9.2
Extruded	Depressed grains	48.1	30.6	1.7	19.6
	Prominent grains	62.6	26.6	0.3	10.5

For longer times or higher temperatures, a considerable increase of the oxidation rate occurs when the single-phase transforms into a two-phase zone and α_2-Ti_3Al is underlying the scale.[29] The latter results in formation of a nonprotective mixed alumina–rutile scale, with development of alumina at Ti–Al–O compound zones and rutile on α_2-Ti_3Al regions. The high Ti activity, high oxygen solubility, and rapid oxygen diffusion should account for the preferential oxidation of α_2-Ti_3Al to rutile.[23,25,53]

It is also expected that these processes can affect oxidation in air because of differences in scale thickness observed for the different alloys. The stages at which this transformation can occur differ significantly depending on the microstructure of the alloy. In γ-TiAl single-phase regions, the transformation follows the stages described above. However, in γ/α_2 duplex regions, the Al-depleted zone consists of two phases: Ti–Al–O phase formed on γ-TiAl and α_2-Ti_3Al lamellae present in the alloy. The fine lamellar structure of cast material led to thinner thickness of the scale than that found in the lamellar grains of extruded material in which α_2-Ti_3Al

plaques are thicker. Therefore, factors like size, distribution and relative amount of α_2-Ti_3Al lamellae strongly affect oxidation rate.

At 900°C, the different stages reported before appeared for shorter oxidation times due to the increase of diffusive processes in the scale and subsurface zone.[26] For instance, the scale developed after 1 hr at 900°C coincided with that observed after 10 hr at 800°C. However, for longer times, a different oxidation behavior to that described before was observed and the thicker scale was developed by the material with a fully lamellar structure. This change of the oxidation behavior could be explained by dissolution of the intermediate-alumina layer and further reprecipitation as particles at the outer part of the rutile scale.[20] When the exposure time increases, the outer rutile scale thickens and, therefore, the oxygen partial pressure close to the alumina barrier can decrease to a critical value for alumina dissolution.

After equal exposure time, the material with the duplex structure showed an almost intact alumina layer with protuberances growing into the rutile scale. The formation of these protuberances suggests that outward Al transport could occur along preferential diffusion paths, which would be related to the differences in the oxide scale developed in the γ/α_2 and γ-TiAl grain regions.

Also noted was the development in the fully lamellar material of a subsurface zone (similar to that observed for longer times at 800°C in extruded material); α_2-Ti_3Al showed significant growth and became larger than the α_2-Ti_3Al lamellae in the matrix. As shown in Fig. 17, oxidation progressed through this phase. Therefore, it would be expected that coarsening of α_2-Ti_3Al would lead to an increase in the oxidation rate.

Concerning the influence of chromium on the oxidation behavior of γ-TiAl-base alloys, it is generally accepted that it produces a detrimental effect at low concentrations ($< 5\%$),[2–7,10] although it is also possible to find contradictory results in the literature.[2] Discrepancies could be associated with several factors, such as different Cr-content, atmosphere, temperature range, or microstructure, since all of these can modify the oxidation behavior of γ-TiAl-base alloys. Accelerated oxide growth rate produced by chromium additions has been explained by the incorporation of Cr^{3+} in the rutile lattice. This is accompanied by an increase of oxygen ion vacancies and/or titanium interstitials that would accelerate the rutile growth rate.[3,4,6,7,15] However, the sequence of oxide formation found in the present work seems to disagree with this hypothesis. At 700°C, a scale without a continuous rutile layer controlling oxidation developed. Moreover, oxidation at temperatures above 800°C developed an outer rutile layer, but Cr mappings revealed only the presence of this element in the inner, rutile–alumina mixed-oxide layer. The enhanced oxidation rate induced by Cr additions

could be associated with a modification on the diffusion process, which would allow faster development of the two-phase, Al-depleted subsurface zone. This effect could explain the formation of a thicker scale in γ-TiAl regions in the duplex material after longer exposure times at 800 and 900°C.

In the present study, it was found that the effect of chromium depended on the microstructure of the alloy. Isolated investigation of these alloys could lead to diverse interpretations of the chromium effect if the microstructure of the alloy is not considered. The influence of alloy microstructure on the oxidation behavior could be extended to other alloying additions. Therefore, it is necessary to consider the effect of this variable to determine the authentic influence of an alloying element and not the effect produced by the microstructure of the alloy.

Scale detachment from both types of microstructures was observed at the higher temperatures. For the fully lamellar material, spalling commenced after 20 and 1 hr at 800 and 900°C, respectively, for the duplex material it was observed only at 900°C after 50 hr exposure. Spallation when the scale reached a critical value, around 10 μm, as reported by Becker *et al.*,[54] was observed only in material with a uniform scale, i.e., the material with a fully lamellar structure. The duplex microstructure of the extruded material leads to an irregular scale, with thin and thick regions, more resistant to spalling. In some way, the stresses generated by both the scale–alloy thermal expansion mismatch and oxide growth can be more easily relaxed by a substrate with softer γ single-phase regions. Preferential nodule formation over γ regions could be the result of increased Ti and Al diffusion rates because of large compressive stresses operating in these regions.

Finally, it is interesting to remark that an adequate control of both alloying elements and microstructure might lead to the design of highly resistant TiAl-base alloys with technical significance for the real world. The present work has shown good adherence of a thick oxide scale ($> 100\ \mu$m after 160 hr at 800°C). Combining the positive effect of the microstructure with an appropriate selection of alloying elements (to reduce the growth rate of the oxide scale) could imply a significant lifetime extension of TiAl-base alloys.

CONCLUSIONS

From the present study, the following conclusions can be established.

- The oxidation behavior of Ti–46Al–1Cr–0.2Si alloy depends on the type of microstructure. The homogeneous lamellar microstructure of as-cast alloy results in a uniform scale more protective than the nonuniform scale developed in duplex microstructure of extruded material. Thus, the size

and distribution of α_2-Ti_3Al can determine the oxidation rate and type of scale developed.

- The oxidation process is probably controlled by the different process occurring in the Al-depleted subsurface. The oxidation rate increases when α_2-Ti_3Al is present in addition to the Ti–Al–O compound because of the higher reactivity of this phase with oxygen to form rutile. For the lamellar, cast alloy, this two-phase zone appears in the entire Al-depleted zone from the initial stages of oxidation and a homogeneous scale is formed. For the extruded material, however, the Al-depleted zone initially consists of two phases only in the lamellar grain regions. In the γ-TaAl single-phase regions, longer times and/or higher temperatures are required for the formation of the two-phase zone. Consequently, an irregular scale with thin and thick regions is developed in the extruded material.
- Nonuniform scales on extruded material have a high resistance to spallation. In some way, the stresses generated by both thermal-expansion mismatch between the scale and alloy and oxide growth can be more easily released by a substrate with softer γ single-phase regions.

ACKNOWLEDGMENTS

The authors acknowledge V. A. C. Haanappel and J. L. González Carrasco for critical reading of the paper.

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